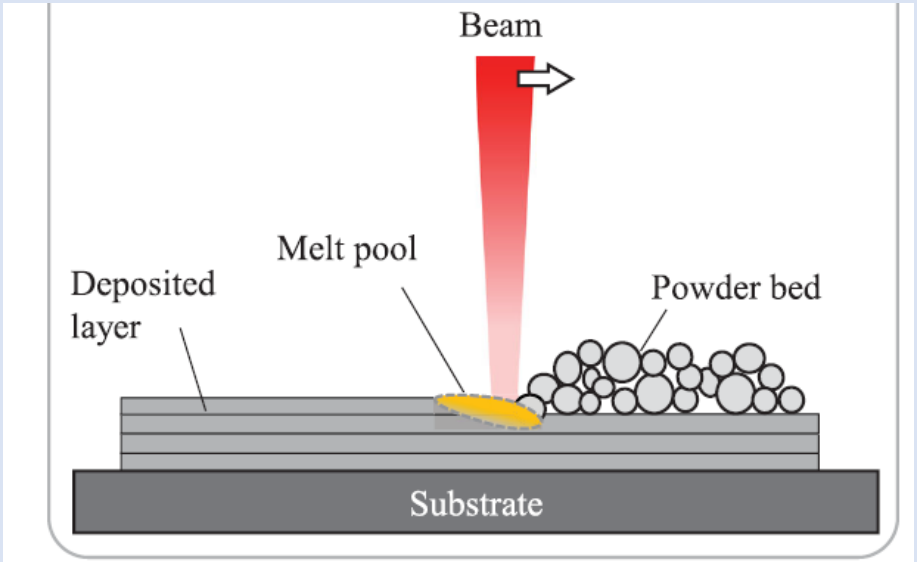
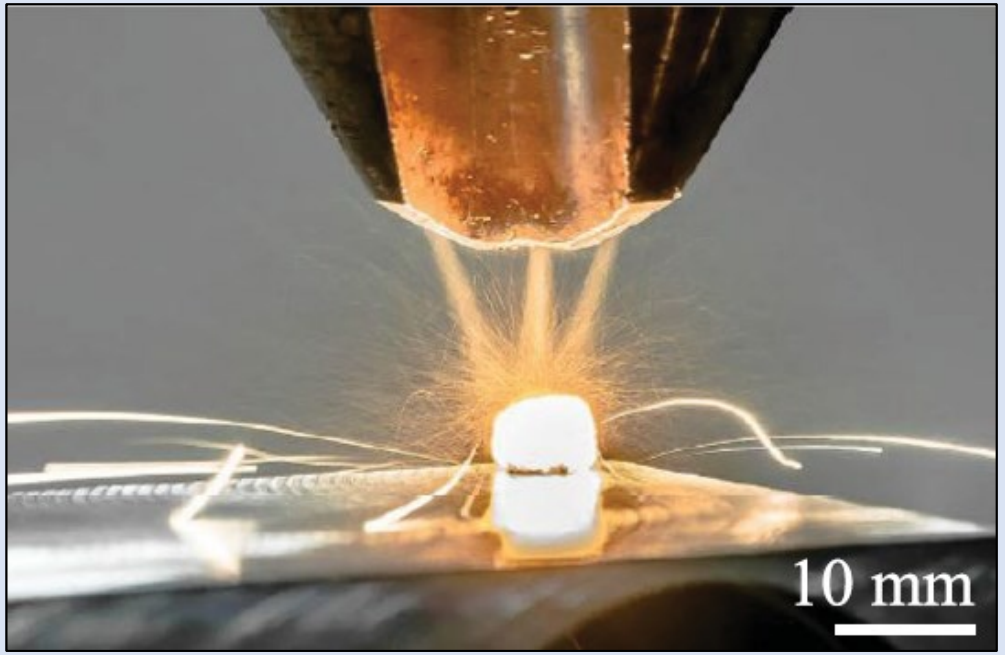
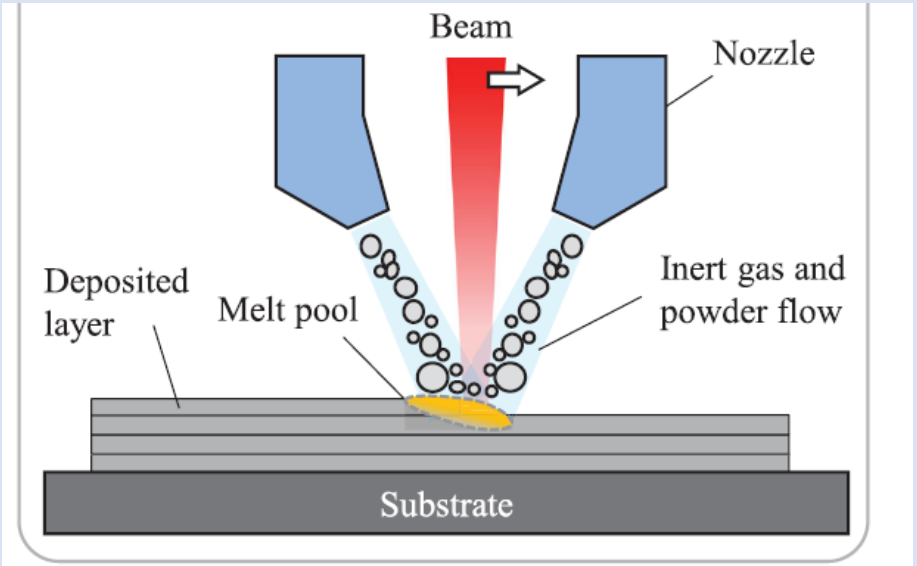
Powder Bed Fusion (PBF)





Directed Energy Deposition (DED)





Powder Bed Fusion (PBF)

Electron Beam Melting	EBM
Vat Photopolymerization	VPP
Binder Jetting (glue)	BJ
Material Jetting	MJ
Sheet Lamination (cut)	SHL

Classificações A.M.:

- Por fonte de energia
- Por tipo de material

Directed Energy Deposition (DED - ARC)



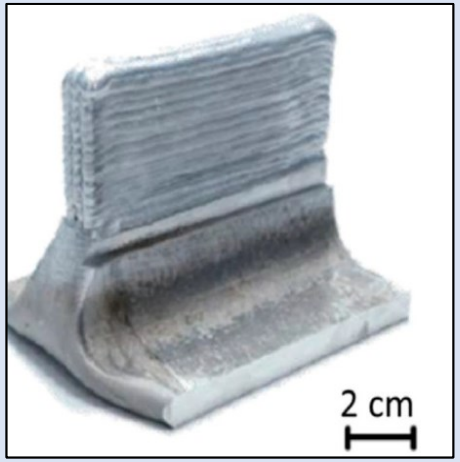
GTAW

Wire Arc

WAAM

Wire DED

Arco Voltaico



Modelling and simulation of metal additive manufacturing processes with particle methods: A review https://doi.org/10.1080/17452759.2023.2274494

Para que modelar?

PROBLEMAS A.M.

1- Orientação – intensidade

2- Descontínuidades (*balling*)

3- Keyholes (espaços vazios - vapor)

4- Degradação química e oxidação

5- Partículas residuais

6- Trincas

7- Delaminação / Distorção

8- Acabamento superficial

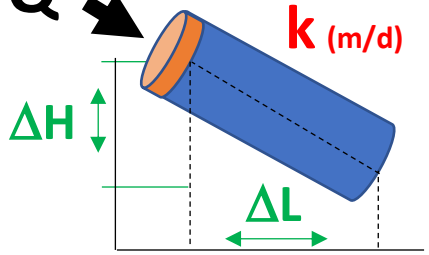
9- Porosidade

- **Racionalizar o problema**
- **Criar hipóteses**
- **... entender a base de dados**
- **Controlar o processo de fabricação**

Grande quantidade de variáveis !

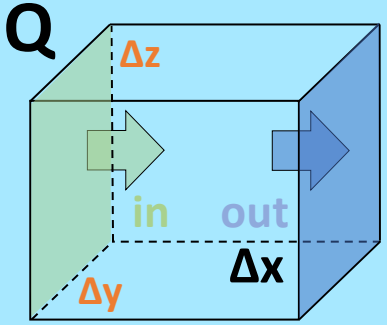
Contexto

Darcy ~ Fourier



$$\frac{Q}{A_{\varnothing}} = q = k\frac{\Delta H}{\Delta L}$$ H_2O

$$\dot{Q} = (-kA\,\Delta T)/L$$ Calor



$$\frac{\partial q_x}{\partial x}\,\Delta x\,\Delta y\Delta z$$

infinitesimal

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} + \frac{\partial^2 h}{\partial z^2} = Q + PA$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial h}{\partial x}\right)$$

Laplace

$$\nabla^2 h$$

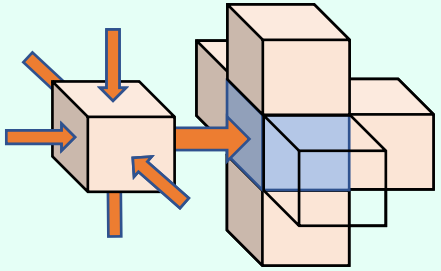
nabla

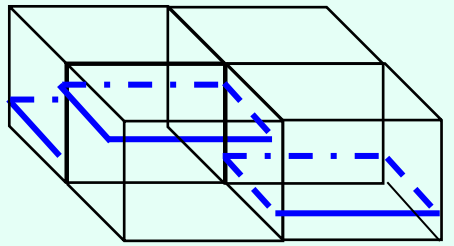
= fontes
(-) sumidouros

MODFLOW

O que é um modelo?

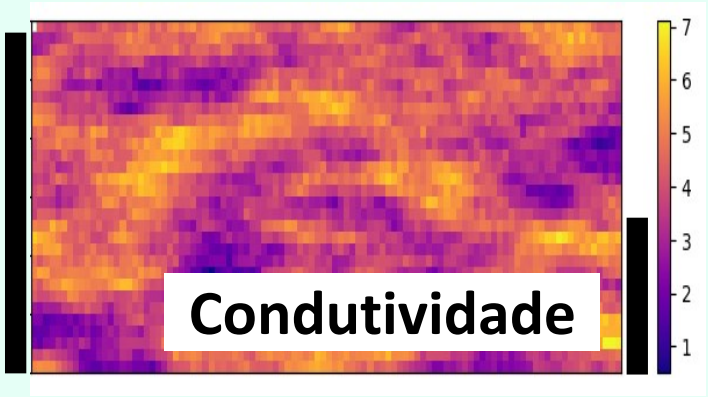
numérico

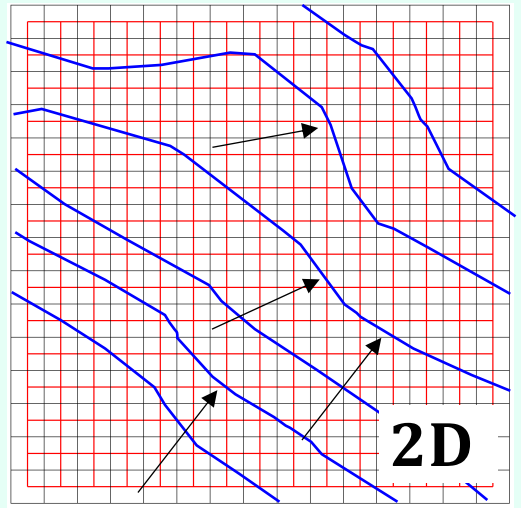




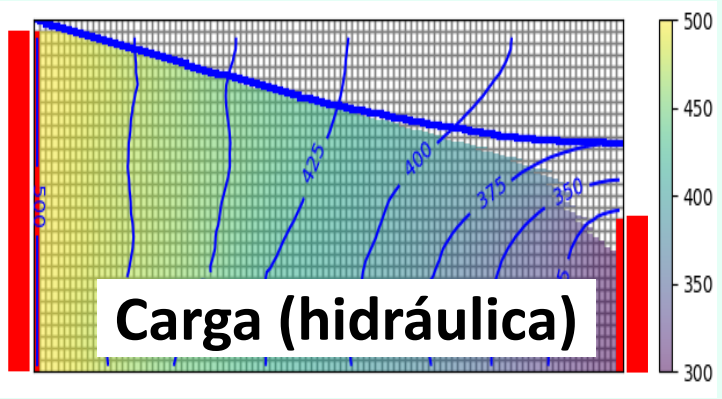
H_n

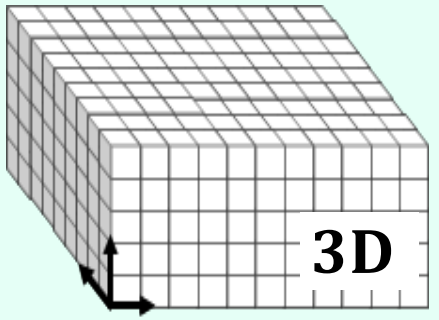
Premissas





Resultados





Excel

https://betami2022.wixsite.com/betami/home-2/t01-spatial-variances/02-inversion/03-iteractive-solution

"pausa"

Otimização Topográfica

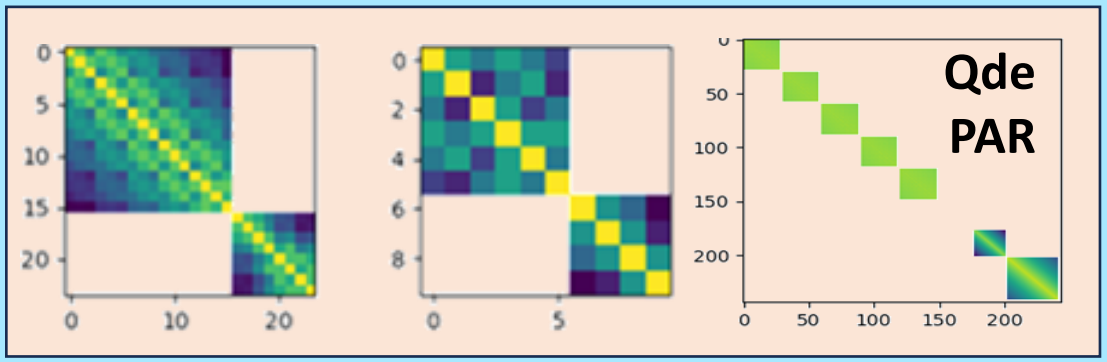
$$\frac{\partial u}{\partial t} + \nabla.(u\psi) - \nabla.(\Gamma \nabla \psi) = S_\psi$$

convecção — difusão — fontes

$$\begin{bmatrix} * & * & & * \\ * & * & * & \\ & * & * & * \\ * & & * & * \end{bmatrix} * \begin{bmatrix} \psi \end{bmatrix} = \begin{bmatrix} * \\ * \\ * \\ * \end{bmatrix}$$

A * **ψ** = **b**

Modelo — **OBS** ⟷ **PAR**



Notação Diferencial

Notação Matricial

CFD

Inversão

$$\psi = A^{-1} b$$

Parameter ESTimation software — **PEST**

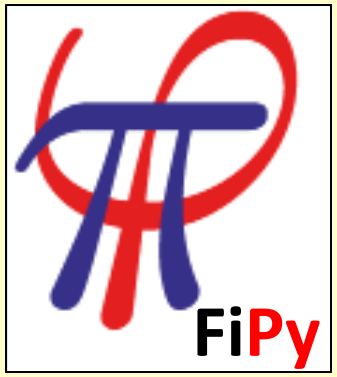
Cálculo Bayesiano

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Probabilidade condicional

Model matrix
Design matrix
Coefficient matrix
Covariance matrix
Regressor matrix
Asymmetric matrix
Sparce Z matrix

Abordagem Estocástica








```
==========
 \\      /
  \\    /
   \\  /
    \\/
```

Open

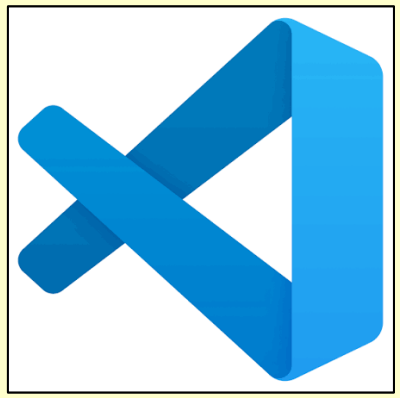
F ield

O peration

A nd

M anipulation





Visual Studio



Jupyter Notebook

O que é um modelo?



✓ **Analítico**

Numérico

```
import numpy as np  ; import matplotlib.pyplot as plt
from fipy import CellVariable, Grid2D, Viewer,
from fipy import TransientTerm, DiffusionTerm
```

```
nx=20  ;ny=20   ;dx=1.;dy=1.  ;L=dx*nx
mesh = Grid2D(dx=dx, dy=dy, nx=nx, ny=ny)
phi  = CellVariable(name="solution
variable",mesh=mesh,value=0.)
```

Grid2D

CellVariable





Linguagem orientada a objetos

https://github.com/grossiRM/CFD_ON/blob/main/02_FiPy

"pausa"

Ecossistema

https://github.com/usnistgov

usnistgov

Overview | Repositories 1.4k | Projects 5 | Packages | People 40

National Institute of Standards and Technology

Department of Commerce

2k followers | Gaithersburg, Md. | https://www.nist.gov

FiPy

https://github.com/new

Create a new repository

Repositories contain a project's files and version history.

Required fields are marked with an asterisk (*).

1 General

Owner * grossiRM / Repository name * new

new is available.

https://github.com/grossiRM/CFD_ON

CFD_ON Public

main | 1 Branch | 0 Tags

grossiRM 14

01_DeGroot	02
02_FiPy	07
03_OBJ_FUNC	02
04_OpenFoam	14
05_Linkedin_Posts	08
06_PyLops	04
_aux	04
.gitignore	10
README.md	06
desktop.ini	02
w_14_control_.txt	14

Exemplos:

02__ https://pages.nist.gov/fipy/en/latest/generated/examples.diffusion.mesh20x20.html __

02__ https://stackoverflow.com/a/76601829 __ Gaussian surface

```
In [1]: import numpy as np  ; import matplotlib.pyplot as plt                          ; import fipy as fp
        from fipy import CellVariable, Grid2D, Viewer, TransientTerm, DiffusionTerm   ; from fipy.tools import numerix

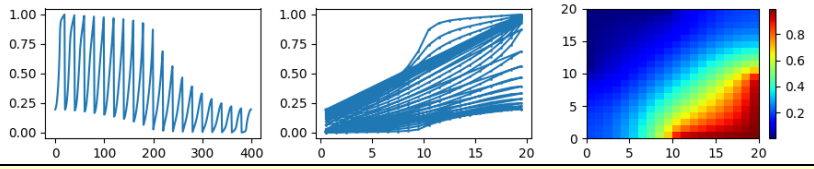
In [ ]: %%capture
        nx=20 ;ny=20   ;dx=1.;dy=1.  ;L=dx*nx     ;mesh = Grid2D(dx=dx, dy=dy, nx=nx, ny=ny)   ; phi=CellVariable(name="solution variable",mesh=mesh,value=0.)

        D = 1.           ; eq = TransientTerm() == DiffusionTerm(coeff=D)                    ; valueTopLeft=0;valueBottomRight=1              ; X, Y = mesh.faceCenters
        facesTopLeft     = ((mesh.facesLeft  & (Y>L/2)) | (mesh.facesTop    & (X<L/2)))     ; phi.constrain(valueTopLeft     ,facesTopLeft)     # Neumann BC
        facesBottomRight = ((mesh.facesRight & (Y<L/2)) | (mesh.facesBottom & (X>L/2)))     ; phi.constrain(valueBottomRight ,facesBottomRight) ; print(len(phi.value))

In [3]: timeStepDuration = 10 * 0.9 * dx**2 / (2 * D)       ; steps = 10
        for step in range(steps):   eq.solve(var=phi,dt=timeStepDuration)

In [4]: YY=phi.value  ;XX=np.linspace(0,len(YY)-1,num=len(YY))                          ; XA = mesh.cellCenters[0]
        xx = mesh.cellCenters[0].value.reshape(nx, ny)   ; yy = mesh.cellCenters[1].value.reshape(nx, ny)   ; zz = phi.value  ; ZZ = zz.reshape(nx, ny)

        fig, (ax1, ax2, ax3) = plt.subplots(1, 3,figsize=(9, 2))                        # ; print(valueBottomRight)
        ax1.plot(XX,YY) ; ax2.plot(XA.value, phi.value, 'o-', markersize=1)        ;WW = ax3.pcolormesh(xx, yy, ZZ,cmap='jet') ;  cbar = plt.colorbar(WW)  ; plt.tight_layout() ; plt.show()
```

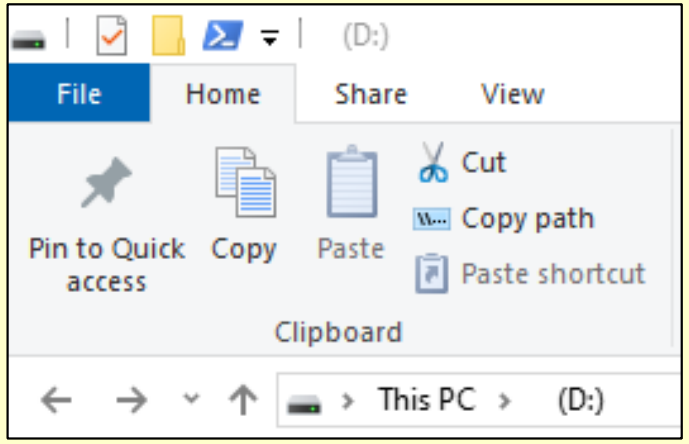


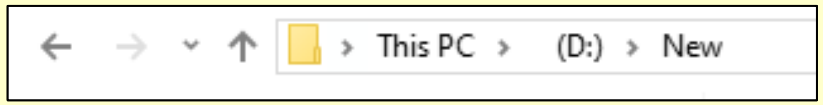
https://github.com/grossiRM/CFD_ON/blob/main/02_FiPy/02_diffusion.mesh20x20.ipynb

Ecossistema

"pausa"

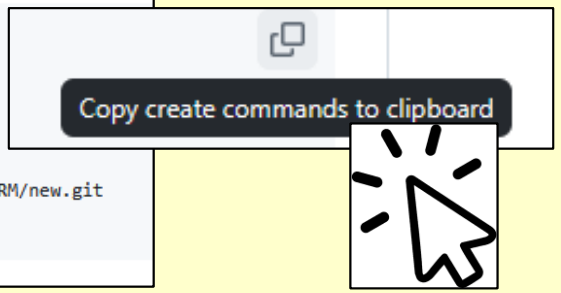
https://github.com/new

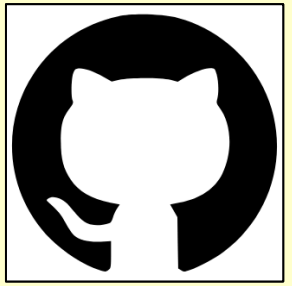




...or create a new repository on the command line

```
echo "# new" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/grossiRM/new.git
git push -u origin main
```

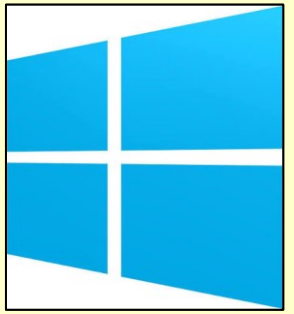




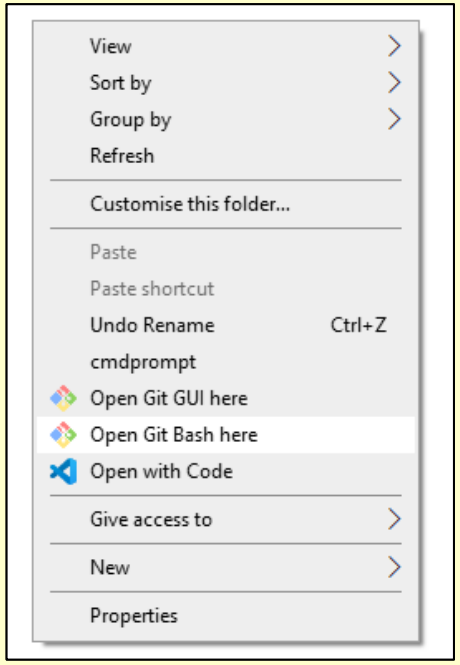
Git

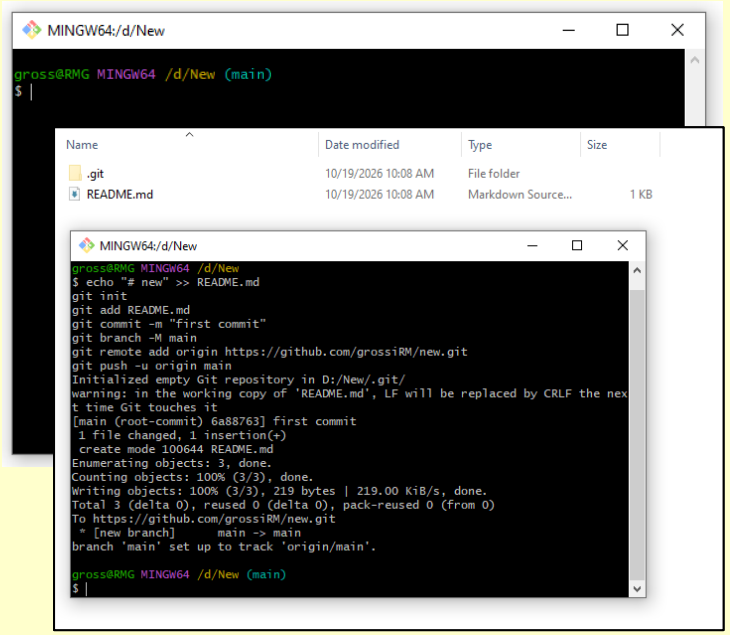


Explorer

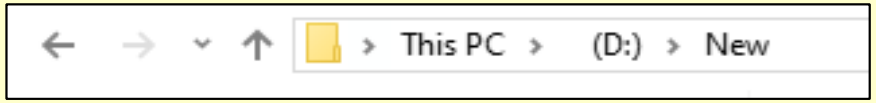


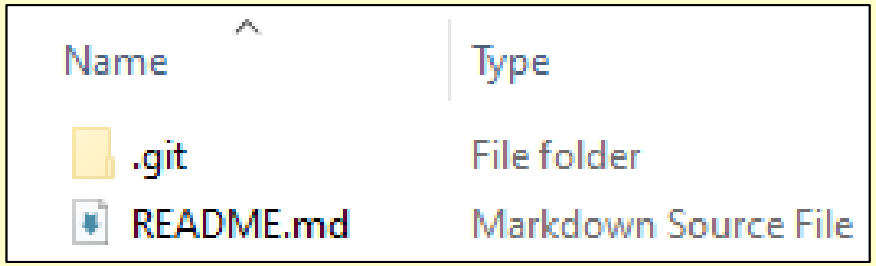
Windows






```
echo "# new" >> README.md
git initgit add README.md
git commit -m "first commit“
git branch -M main
git remote add origin https://github.com/grossiRM/new.git
git push -u origin main
```






```
$ git add .
$ git commit -m “001"
$ git push origin
```

https://github.com/grossiRM/AM_FEG

- __pycache__
- builders
- representations
- topologies
- __init__.py
- abstractMesh.py
- cylindricalGrid1D.py
- cylindricalGrid2D.py
- cylindricalNonUniformGrid1D.py
- cylindricalNonUniformGrid2D.py
- cylindricalUniformGrid1D.py
- cylindricalUniformGrid2D.py
- factoryMeshes.py
- gmshMesh.py
- grid1D.py
- grid2D.py
- grid3D.py
- mesh.py
- mesh1D.py
- mesh2D.py
- nonUniformGrid1D.py
- nonUniformGrid2D.py
- nonUniformGrid3D.py
- periodicGrid1D.py
- periodicGrid2D.py
- periodicGrid3D.py
- skewedGrid2D.py
- sphericalNonUniformGrid1D.py
- sphericalUniformGrid1D.py
- test.py
- tri2D.py
- uniformGrid.py
- uniformGrid1D.py
- uniformGrid2D.py
- uniformGrid3D.py

fipy

```
from fipy import Grid1D ;nx=50 ;dx=1. ;mesh =Grid1D(nx=nx,dx=dx)
```

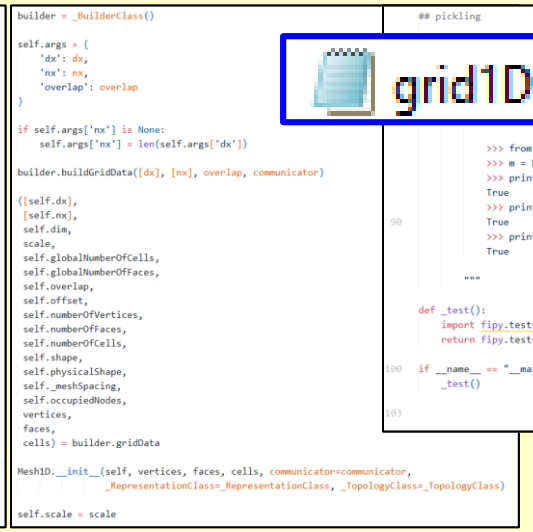
DIFUSÃO

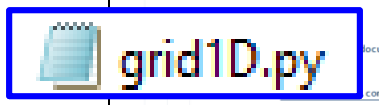
$$\frac{\partial \phi}{\partial t} = D\nabla^2\phi$$

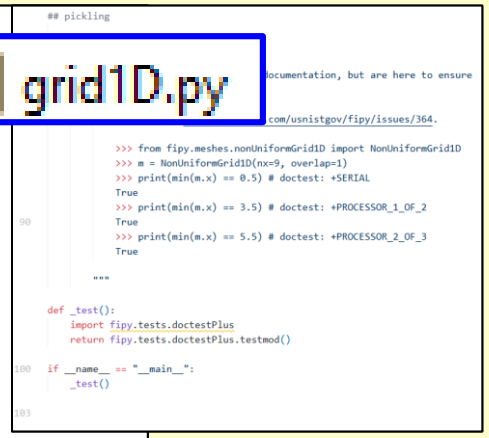
Python





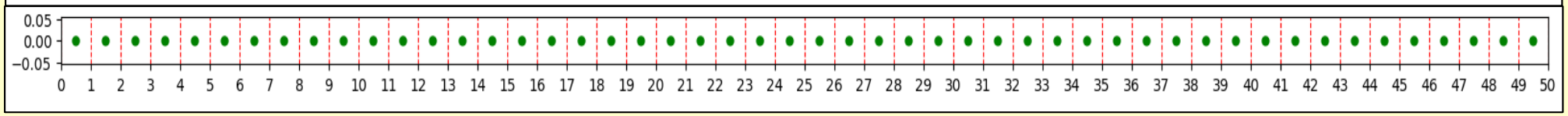




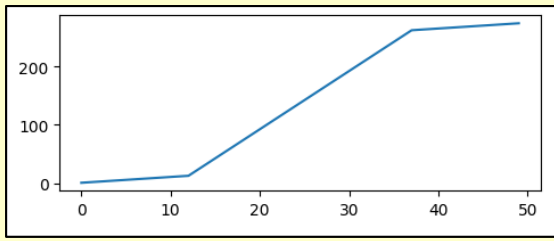


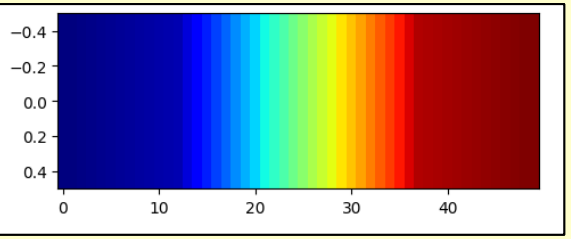
Objeto.py

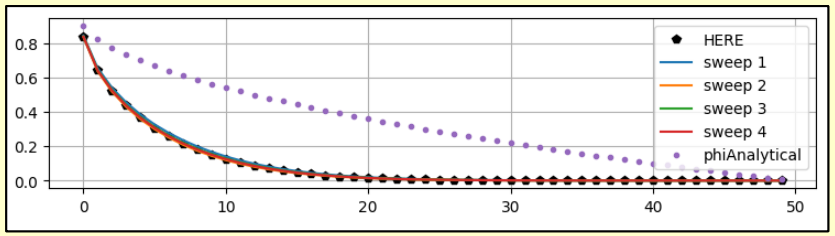
```
plt.figure(figsize=(20, 0.5)) ;FF = mesh.faceCenters()
;X=mesh.faceCenters[0] ;plt.xlim(FF.min(),FF.max());plt.xticks(X)
plt.plot (mesh.cellCenters(),np.zeros_like(mesh.cellCenters()),'o-',markersize=5,color='green')
plt.grid (True, axis='x', linestyle='--', color='red')
```



... Condições iniciais e de contorno, solvers, equação de referência, visualização

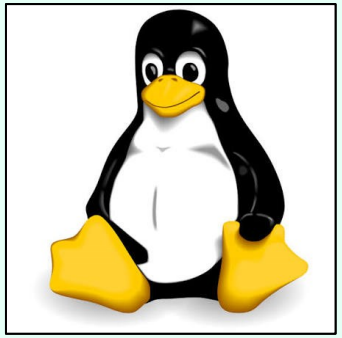






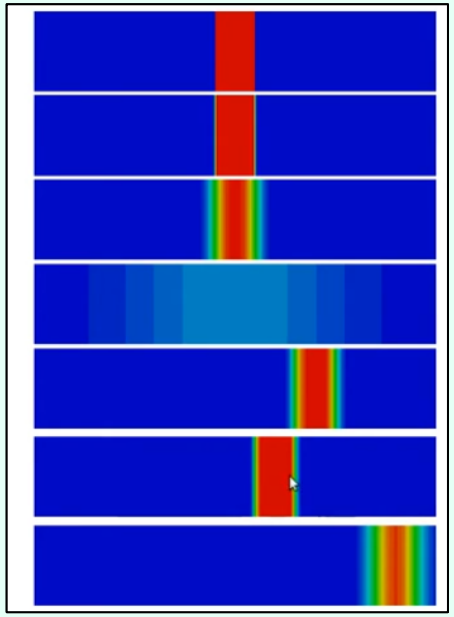
```
==========
 \\      /   F ield
  \\    /    O peration
   \\  /     A nd
    \\/      M anipulation
```

Open



Linux

ubuntu



Convection

Diffusion

Exemplos:

Convection

source

0

$$\frac{\partial \rho \Phi}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \Phi) = \nabla \cdot (D \nabla \Phi) + S_{\Phi}(\Phi)$$

Diffusion

0

$$\frac{\partial \rho \Phi}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \Phi) = \nabla \cdot (D \nabla \Phi) + S_{\Phi}(\Phi)$$

transient

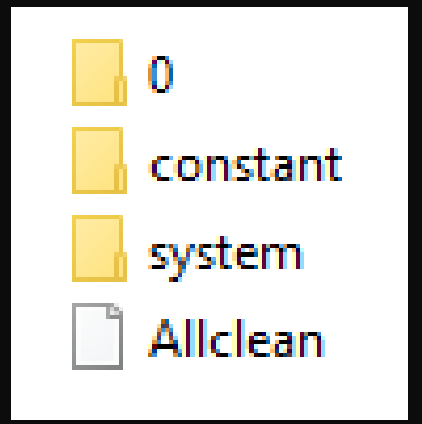
Transport equation:

$$\frac{\partial T}{\partial t} + \nabla \cdot \boldsymbol{u} T = D_T \Delta T$$

0

Diffusion

```
grossi@RMG:~/OpenFOAM/grossi-v2412/run/forwardStep$ tree
.
├── 0
│   ├── T
│   ├── U
│   └── p
├── Allclean
├── constant
│   ├── momentumTransport
│   └── physicalProperties
└── system
    ├── blockMeshDict
    ├── controlDict
    ├── functions
    ├── fvSchemes
    └── fvSolution
```



https://github.com/grossiRM/CFD_ON/tree/main/04_OpenFoam

```
==========
 \\      /  F ield
  \\    /   O peration
   \\  /    A nd
    \\/     M anipulation
```

$$\nabla . \mathbf{V} = 0$$ **Continuidade**

$$\rho \frac{\partial \mathbf{V}}{\partial t} = \rho \mathbf{g} - \nabla \mathbf{p} + \mu \nabla^2 \mathbf{V}$$ **Qde de movimento**

$$\rho C_V \frac{\partial T}{\partial t} = K \nabla^2 T + \Phi$$ **Energia**

Solving PDEs in Python

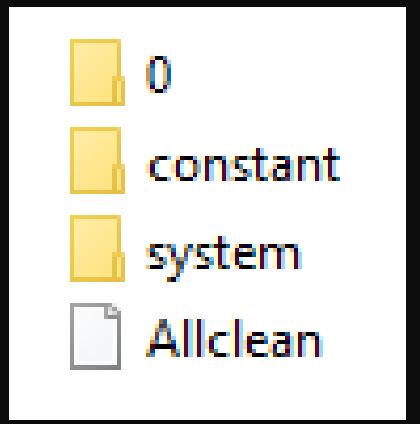


Partial Differential Equations

https://github.com/grossiRM/CFD_ON/blob/main/04_OpenFoam/FEniCS/01b__A%20FEniCS%20Tutorial__ii__.pdf

```
grossi@RMG:~/OpenFOAM/grossi-v2412/run/forwardStep$ tree
.
├── 0
│   ├── T
│   ├── U
│   └── p
├── Allclean
├── constant
│   ├── momentumTransport
│   └── physicalProperties
└── system
    ├── blockMeshDict
    ├── controlDict
    ├── functions
    ├── fvSchemes
    └── fvSolution
```

Exemplos:



https://github.com/grossiRM/CFD_ON/blob/main/04_OpenFoam/03_NOTES-on-CFD_OpenFoam.pdf



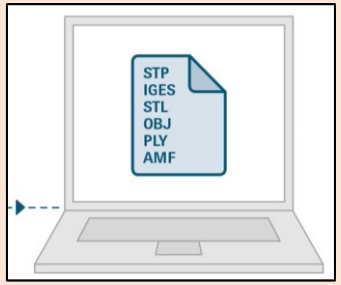
O que modelar?

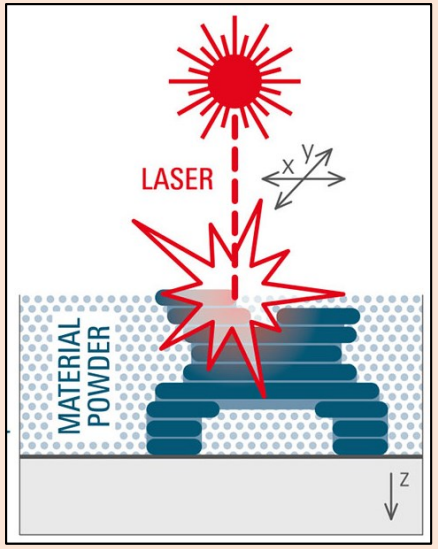
CAD (Computer-Aided Design)

shapes - parametric 3D models (length, width, angles + Rules + History Tree)



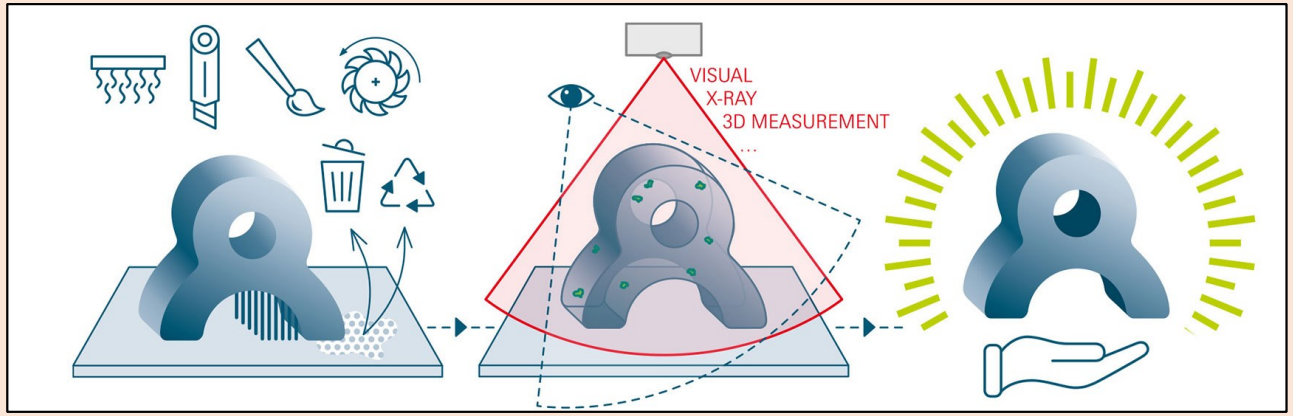
CAE (Computer-Aided Engineering) – data harvesting - structural stresses, deformations, displacement, and failures – **FEA - CFD**





CAM

Computer-Aided Manufacturing



Pós Processamento

Testes, Medições & controle de qualidade

https://formnext.mesago.com/frankfurt/en/industry-insights/am-field-guide/process_in_general.html

Escalas de modelagem

O que modelar?

Macro-scale models

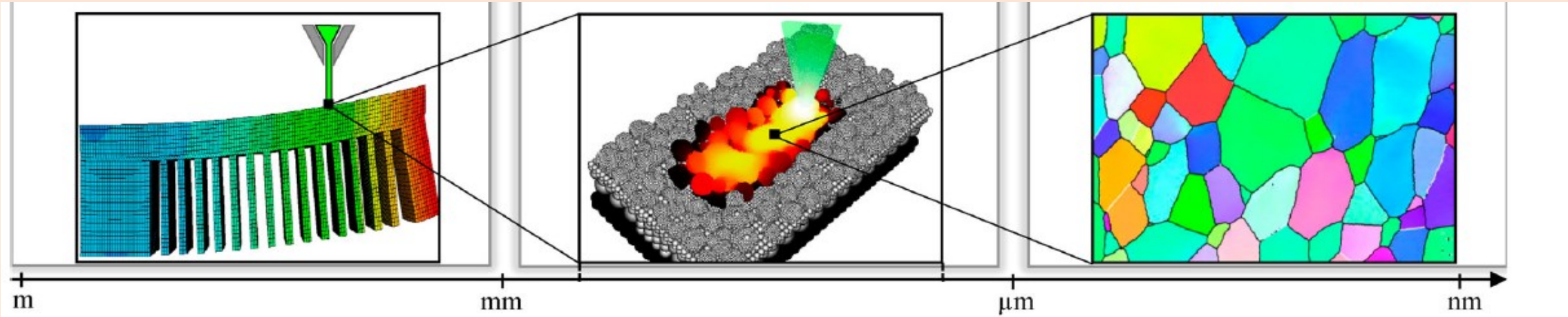
heat flow and structural (stress) deformation & interlayer bonding,

Meso-scale models

pool geometry, bead formation, and dilution.

Micro-scale phase-field

grain growth, microstructural transitions, solidification kinetics



Foco: Distorção, stress residual

Métodos: FEM, FVM

Foco: Fluxo, poça de fusão

Métodos: CFD, SPH, LBM

Foco: Grãos, nucleação, vazios

Métodos: CA, MC, PF

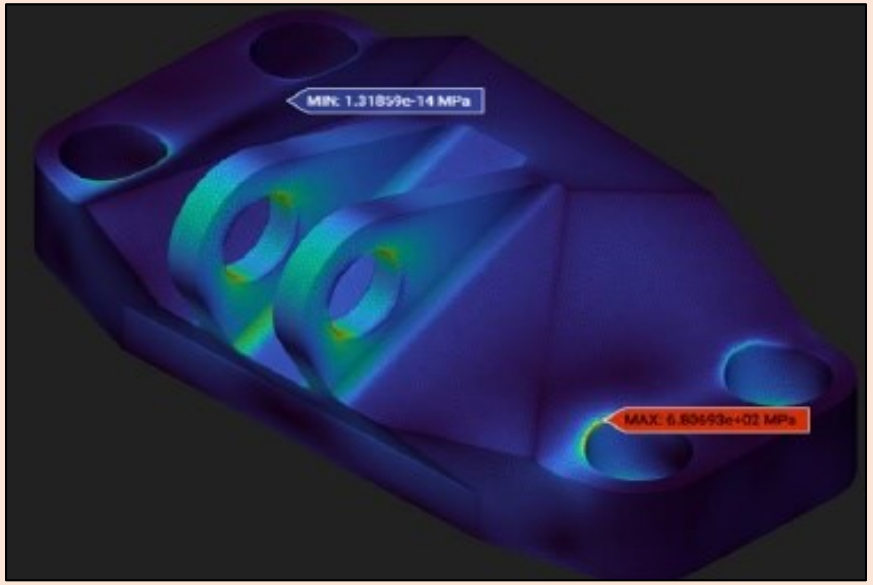
FEA – Análise de Elementos Finitos

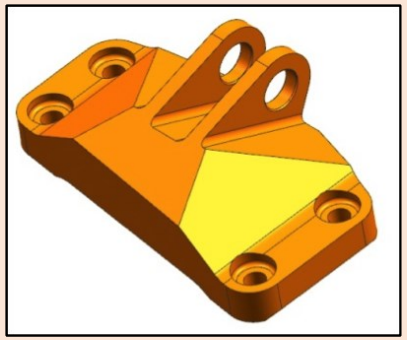
Como modelar?

- Decompõe um objeto 3D complexo em milhares de pequenas partes gerenciáveis ("elementos"); - Calcula tensões estruturais, falhas e deformações.

Stress, Deformação, Vibração, Fadiga

Análise Estrutural



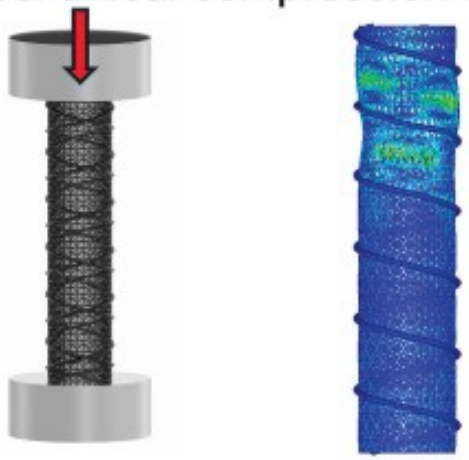


Peça de Suporte

Fluxo, pressão, temperatura, momento, energia

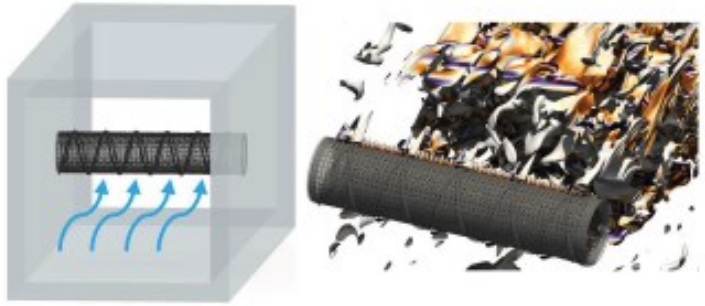
CFD - Fluidodinâmica Computacional

Mechanical compression tests:



FEA

Particle Image Velocimetry (PIV):



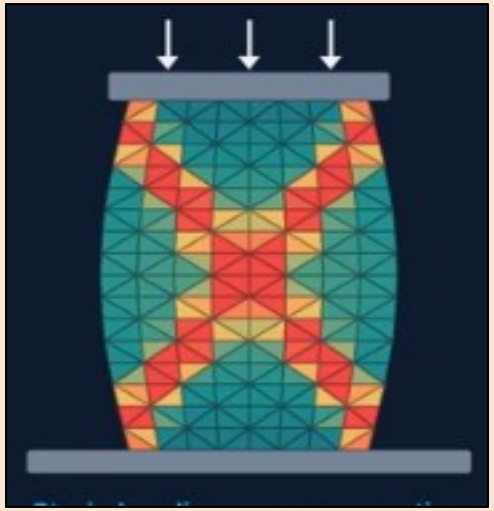
CFD

FSI – *Fluid-Structure Interaction*

Como modelar?

FEM | Finite Element Method

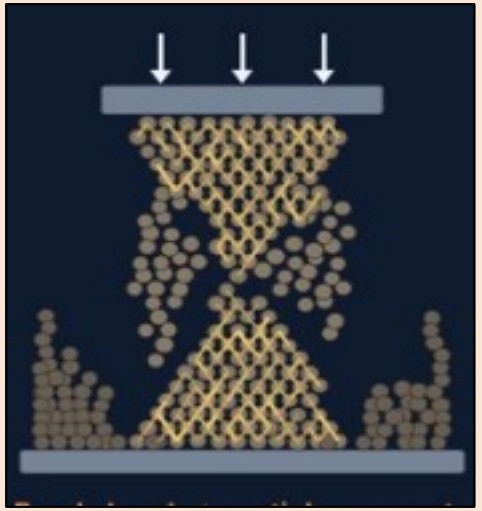
Material contínuo – stress e deformação



DEM | Discrete Element Method

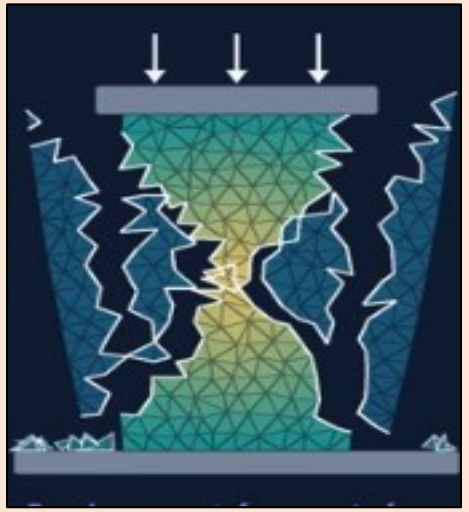
Interação de partículas

Escorregamento, aberturas



FDEM | Finite-Discrete Element Method

Combinação: deformação + interação de blocos = fraturas

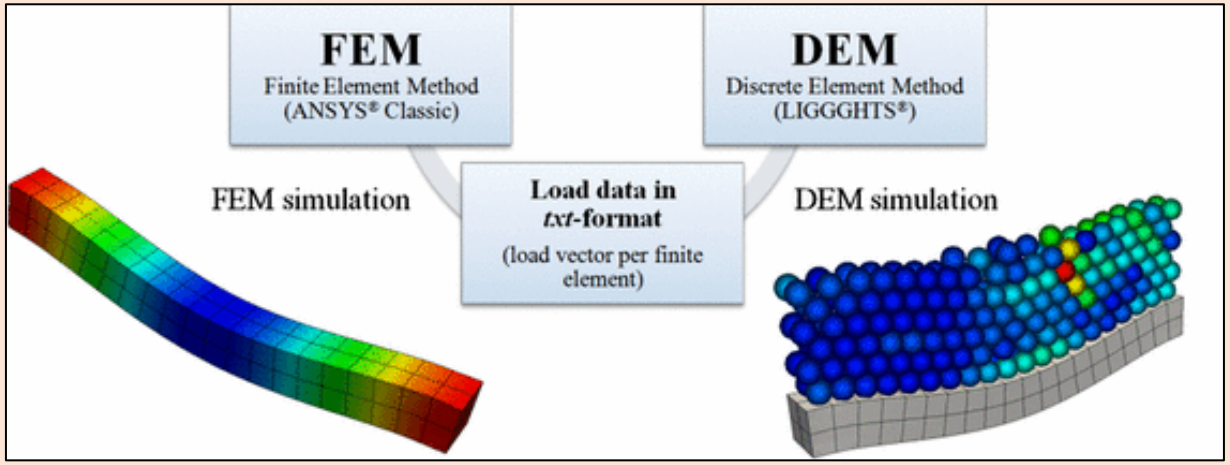


Finite-difference FDM

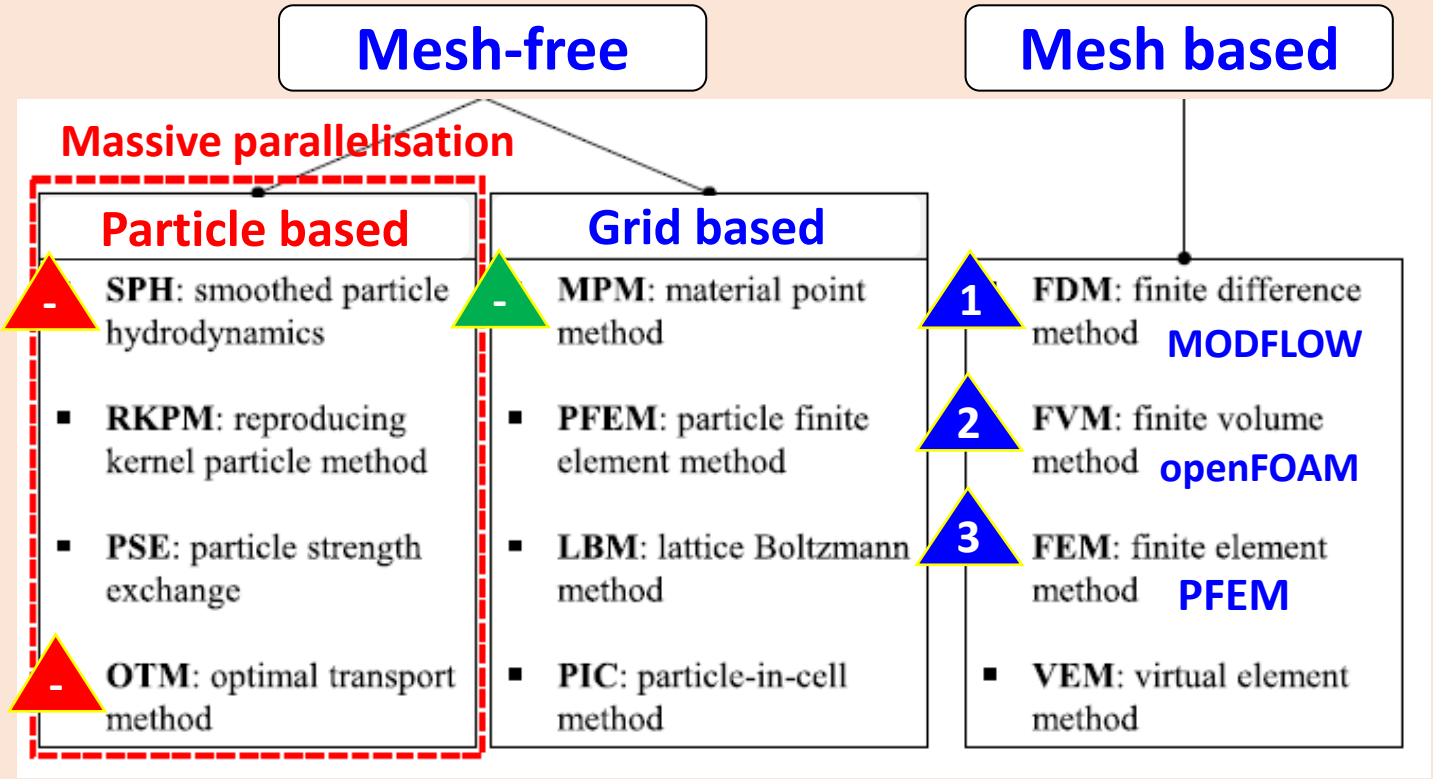
MODFLOW

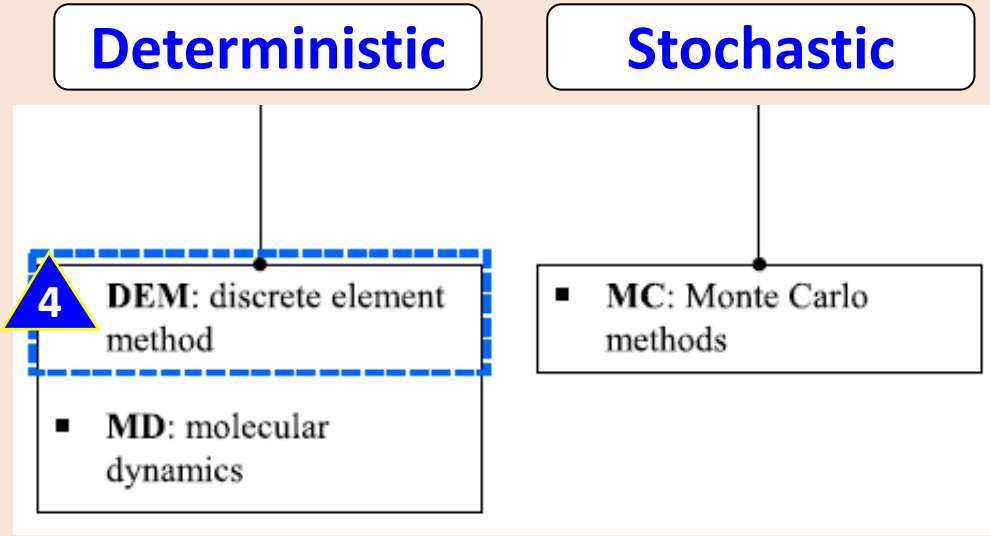
Finite-volume FVM

OpenFOAM, FiPy



Como modelar?





ML algorithms

physics-based **surrogate models**

simulation-informed
closed-loop control feedback
real-time monitoring systems

CFD

Macro

01 - finite element analysis (FEA)

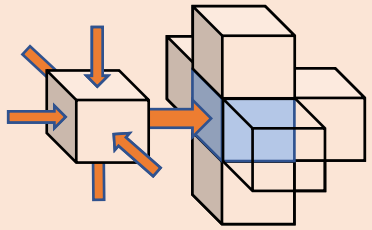
02 - phase-field modeling techniques

Design optimization

heat flow, temperature, and stress evolution during deposition

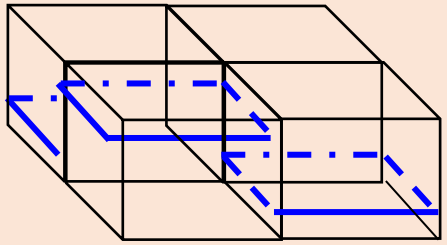
Modelling and simulation of metal additive manufacturing processes with particle methods.

Método de Diferenças finitas (FDM)

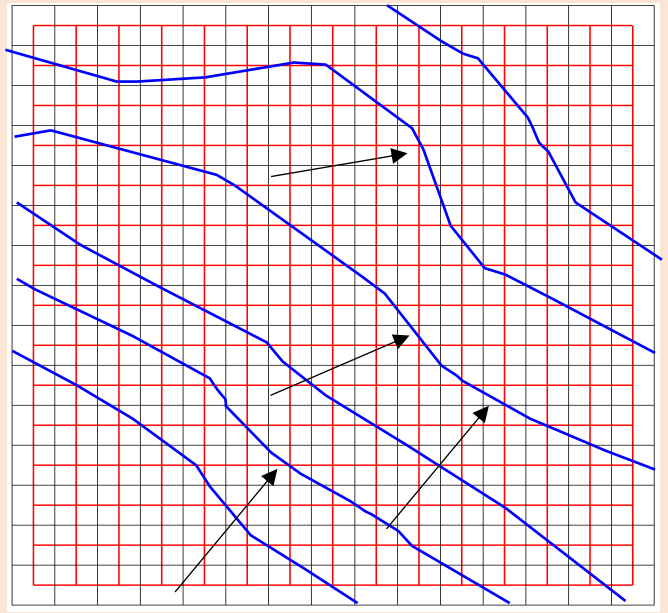


$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

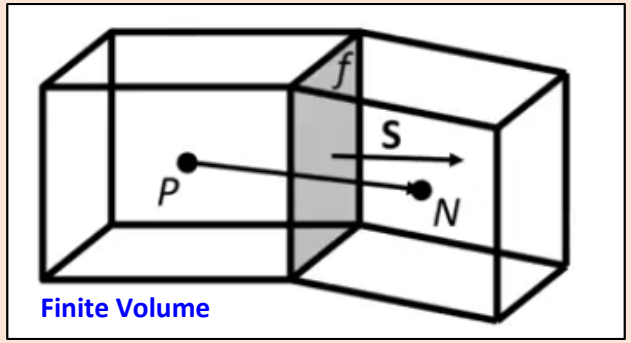
Séries de Taylor



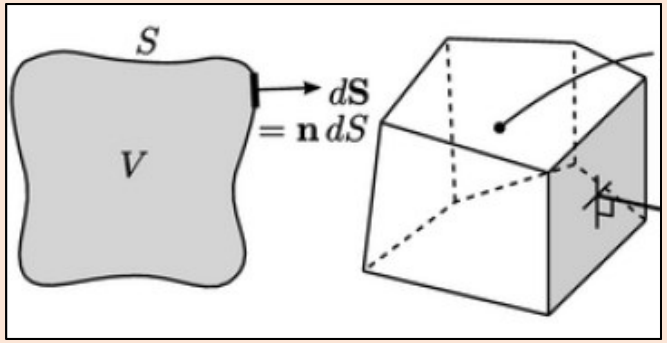
H_n



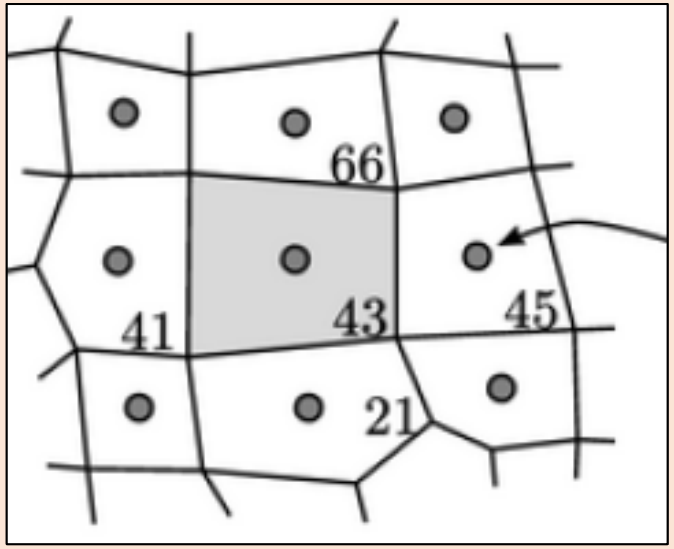
Método de Volumes finitos (FVM)



Finite Volume



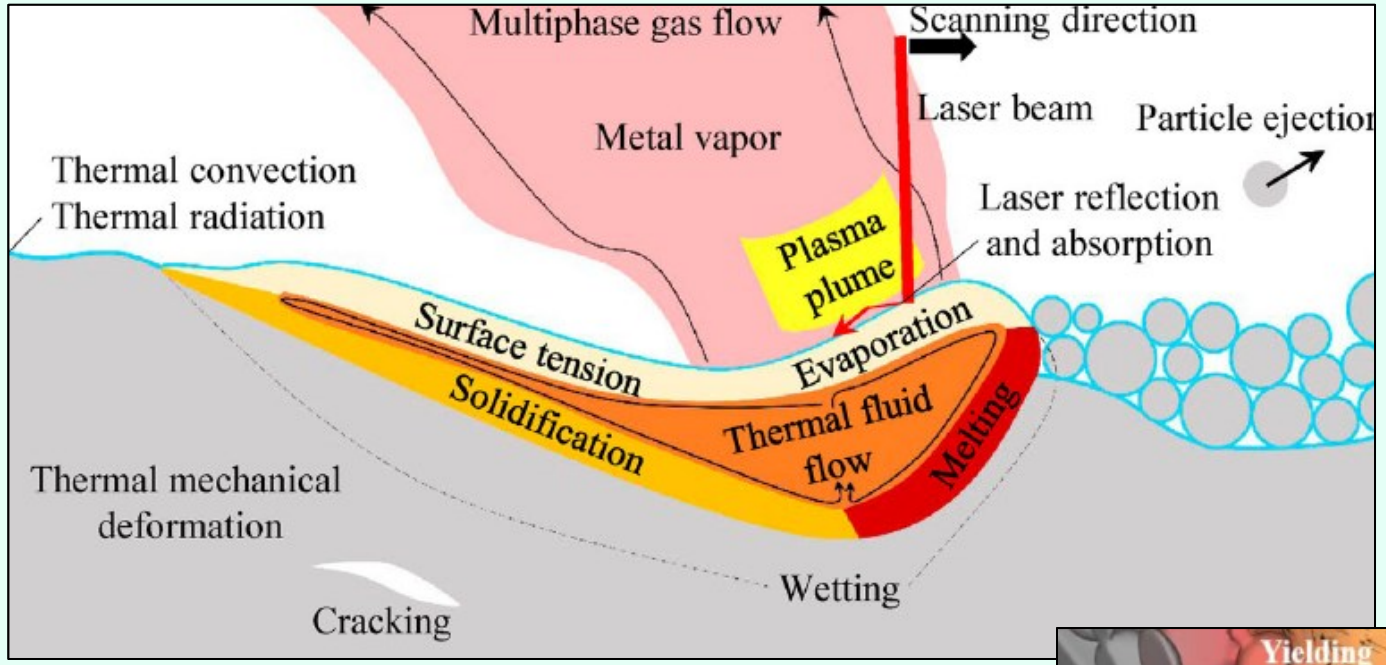
$$\underbrace{\frac{\partial \rho\phi}{\partial t}}_{\text{time derivative}} + \underbrace{\nabla \cdot (\rho \mathbf{U}\phi)}_{\text{convection term}} = \underbrace{\nabla \cdot (\Gamma \nabla \phi)}_{\text{diffusion term}} + \underbrace{S_\phi(\phi)}_{\text{source term}} \quad \textbf{2.3}$$



$$\int_V \nabla \cdot (\rho \mathbf{U}\phi)\, dV = \int_S d\mathbf{S} \cdot (\rho \mathbf{U}\phi)$$

$$\approx \sum_f \mathbf{S} \cdot (\rho \mathbf{U})_f \phi_{f(F,S)}$$

$$= \sum_f F \phi_{f(F,S)} \quad \textbf{2.12}$$



Meso-scale

Mult—physics

Mult—dimensional

$$\nabla . \mathbf{V} = 0$$

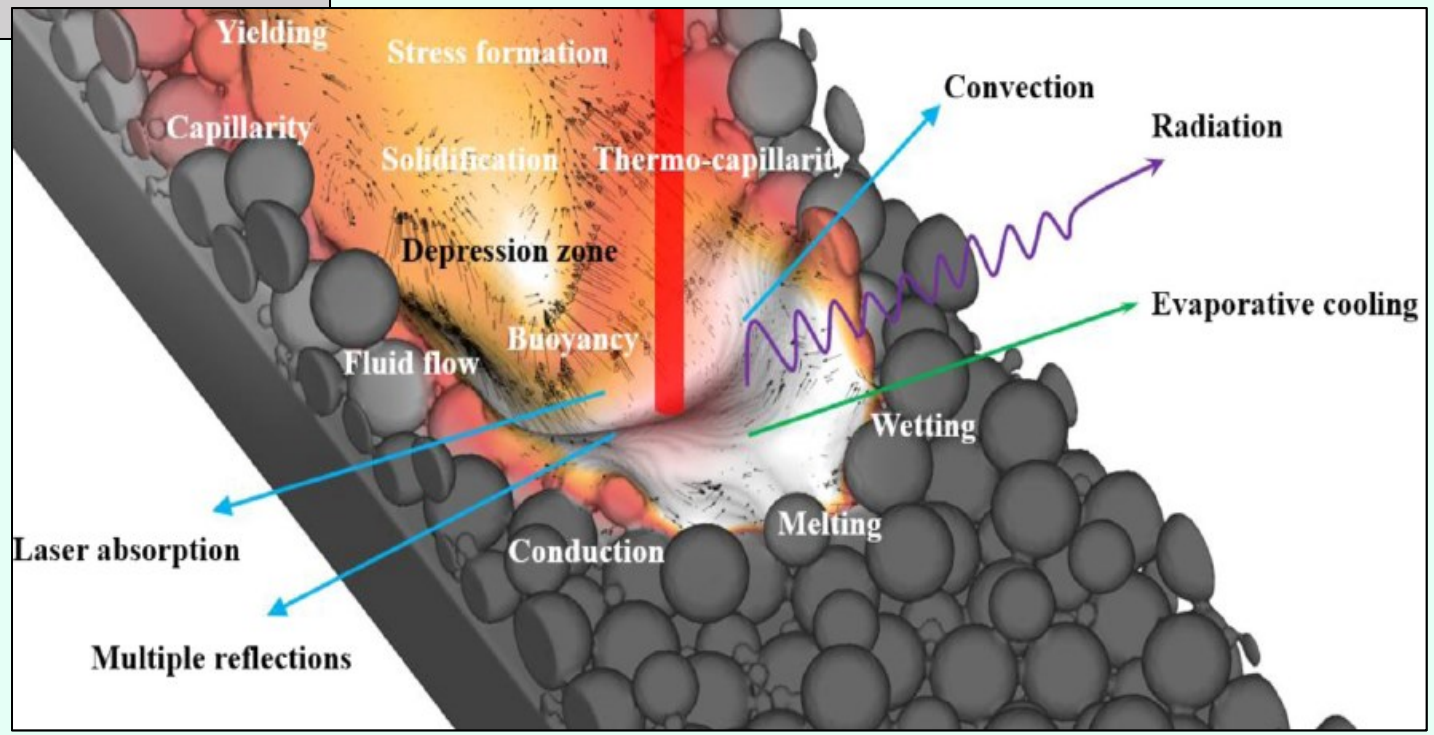
Continuidade

$$\rho \frac{\partial \mathbf{V}}{\partial t} = \rho \mathbf{g} - \nabla \mathbf{p} + \mu \nabla^2 \mathbf{V}$$

Qde de movimento

$$\rho C_V \frac{\partial T}{\partial t} = K \nabla^2 T + \Phi$$

Energia



A review of multi-scale and multi-physics simulations of metal AM processes with focus on modeling strategies

CFD Methodology

https://github.com/DLR-RY/VoFLibrary

CFD

Incompressible, multiphase flow with surface tension and phase change

Volume of fluid method (VOF)

$$\frac{\partial \alpha}{\partial t} + \nabla . (\alpha u) = 0$$

Volume of fraction

$\alpha = V_i / V$

α = 0 (solid)
α = 1 (liquid)
$0 < \alpha < 0$ (interface)

Continuity Equation (mass conservation)

$$\nabla . u = 0$$

Surface tension Marangoni force

Momentum conservation

$$\underbrace{\frac{\partial(\alpha \mathbf{u})}{\partial t}}_{\text{Inertia (rate of change of momentum)}} + \nabla \cdot (\alpha\, \mathbf{u} \otimes \mathbf{u}) = \underbrace{-\nabla p}_{\text{Pressure force}} + \underbrace{\nabla \cdot \tau}_{\text{Viscous force}} + \underbrace{\rho \mathbf{g}}_{\text{Gravity / Body force}} + \underbrace{\sigma \kappa \nabla \alpha + \nabla \sigma \cdot (\mathbf{I} - \mathbf{nn})}_{\text{Surface tension + Marangoni force}}$$

Laser heat input **latent input**

Energy conservation

$$\underbrace{\rho c_p \frac{\partial T}{\partial t}}_{\text{Local (transient) change in temperature}} + \underbrace{\rho c_p\, \mathbf{u} \cdot \nabla T}_{\text{Convective heat transport}} = \underbrace{\nabla \cdot (k \nabla T)}_{\text{Heat conduction}} + \underbrace{Q_{\text{laser}}}_{\text{Laser heat input}} + \underbrace{Q_{\text{phase}}}_{\text{Latent heat (phase change)}}$$

"... the basis of how the simulation solver is modelled and how we can rely on the data that has been predicted ... based on this robust physic equations that are provided "

AM PravaH®

EX: arc welding – analytical solution

Directed Energy Deposition (DED - ARC)

Table 1 Parameters of the welding process

Voltage (V)	Welding current (A)	Speed (mm/s)	Shielding gas	Flow rate (cfh)	Electrode diameter (mm)	Arc length (mm)
14.6	150	1.5	Argon	40	2.4	3

$$\rho c\left(\frac{\partial T}{\partial t}+v\,\nabla T\right)=\nabla.(\mathrm{k}\nabla\mathrm{T})+\dot{Q}$$

Convection Diffusion

new (x) coordinate system

$$\frac{\rho c v}{\mathrm{k}}\frac{\partial \mathrm{T}}{\partial \xi}=\nabla^2 T+\frac{\dot{Q}}{k}$$

$$\xi = y - vt$$

$$T = T_0 + \frac{Q}{2\pi kR} e^{\left(-\frac{\rho c v}{2k}(R+\xi)\right)}$$

$$R=\sqrt{x^2+\xi^2+z^2}$$

Excel

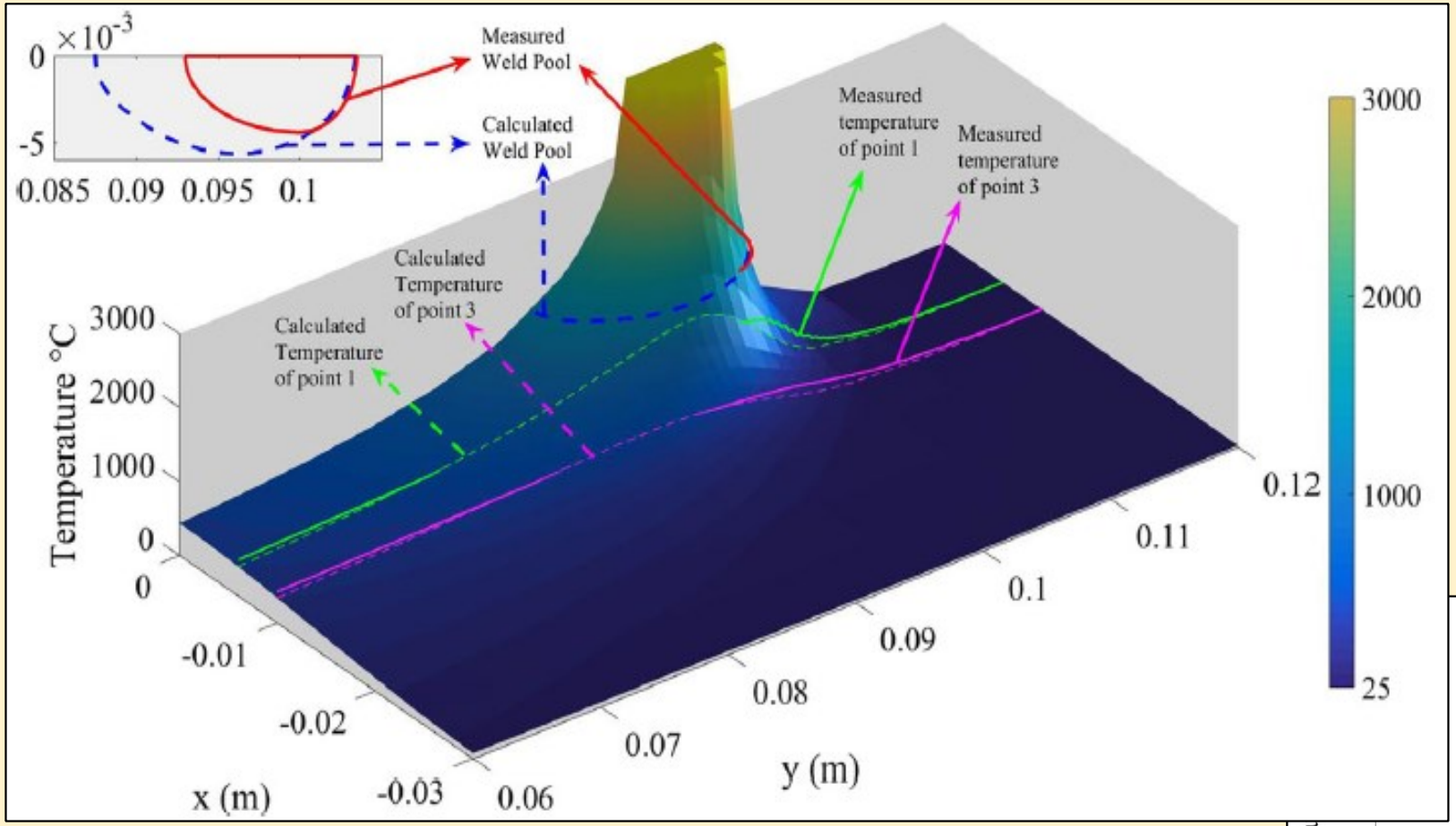


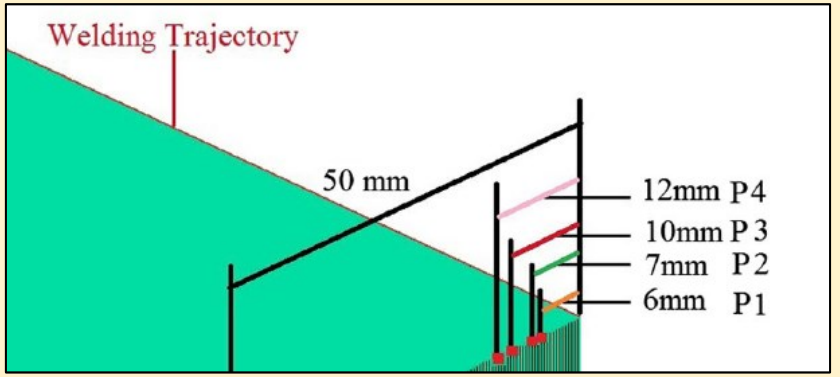
- k Heat conductivity
- R Distance - moving heat source and pt interest
- c Specific heat
- v Welding speed
- ξ Moving coordinate parallel to the welding

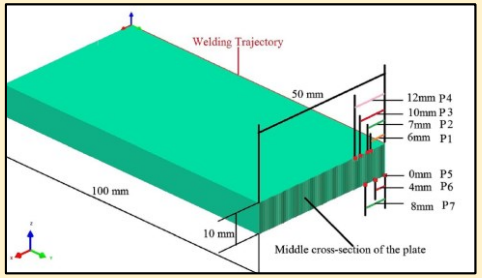
An analytical solution for temperature distribution in fillet arc welding based on an adaptive function Bagher & Norbert Enzinger

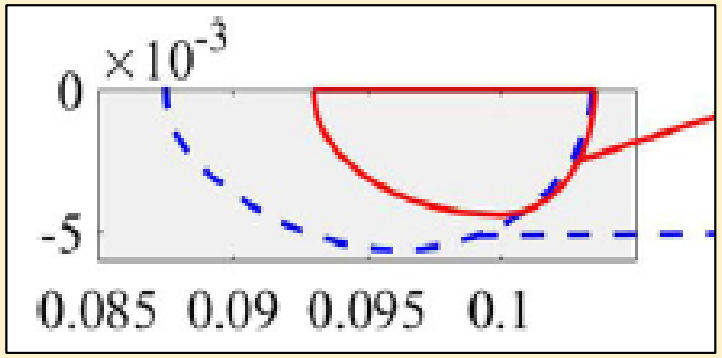
EX: arc welding - analytical solution

Directed Energy Deposition (DED - ARC)

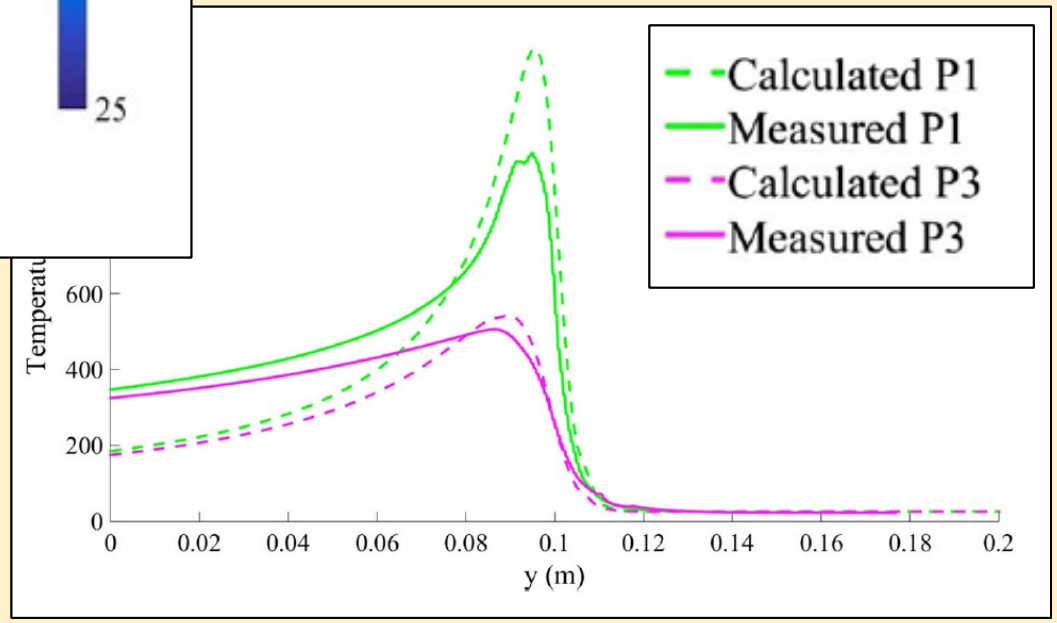








$$T = T_0 + \frac{Q}{2\pi kR} e^{\left(-\frac{\rho c v}{2k}(R+\xi)\right)}$$



Part 01

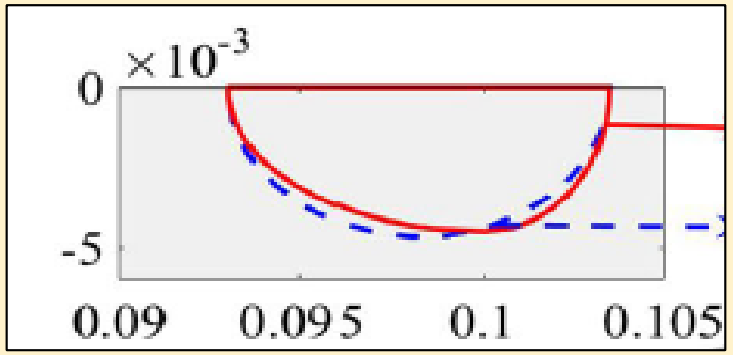
EX:

$$T = T_0 + \frac{Q}{2\pi kR} e^{\left(-\frac{\rho c v}{2k}(R+\xi)\right)}$$

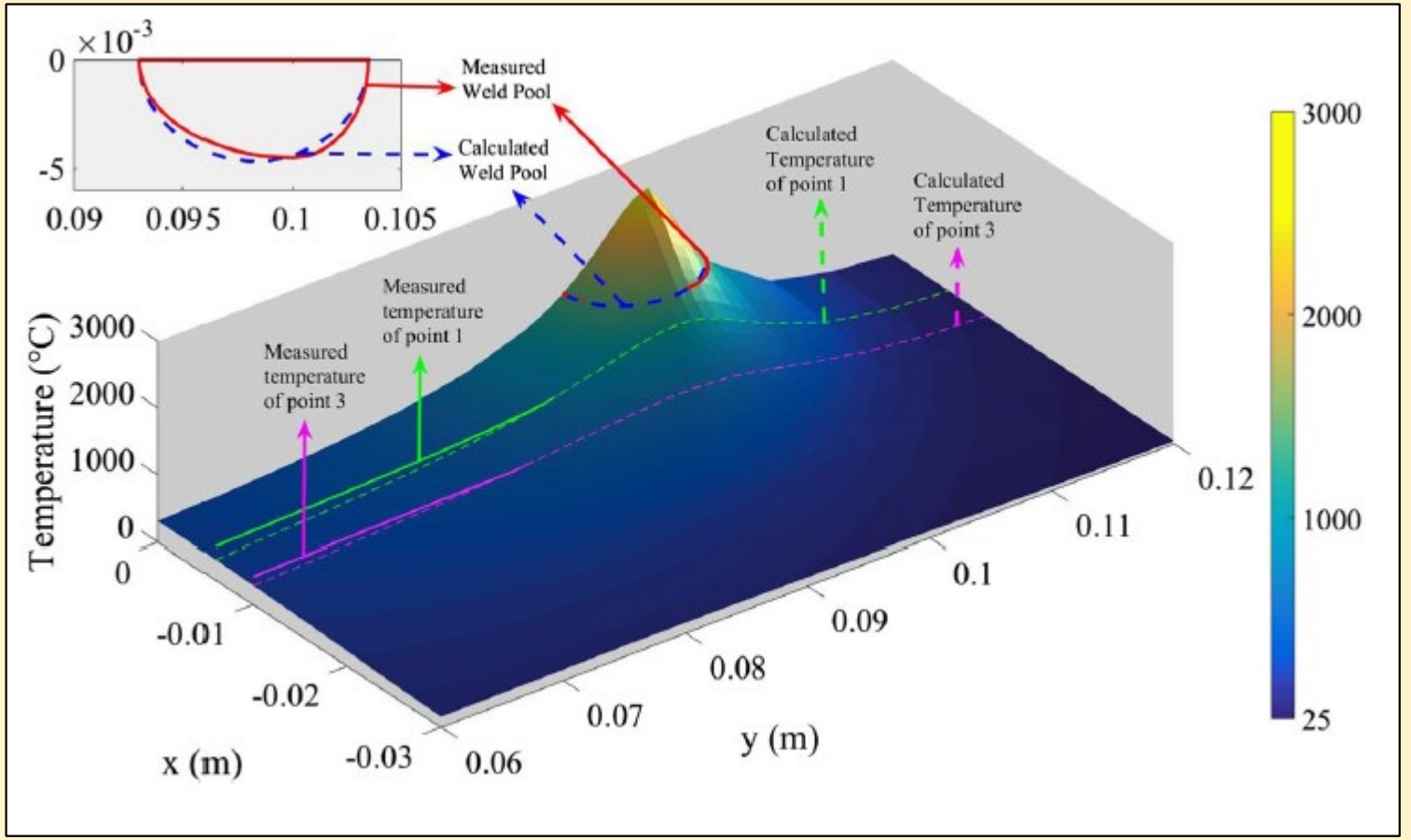
$$R_p^* = \sqrt{\left(\frac{x}{a}\right)^2 + \left(\frac{\xi}{b}\right)^2 + \left(\frac{z}{c}\right)^2 + d^2}$$

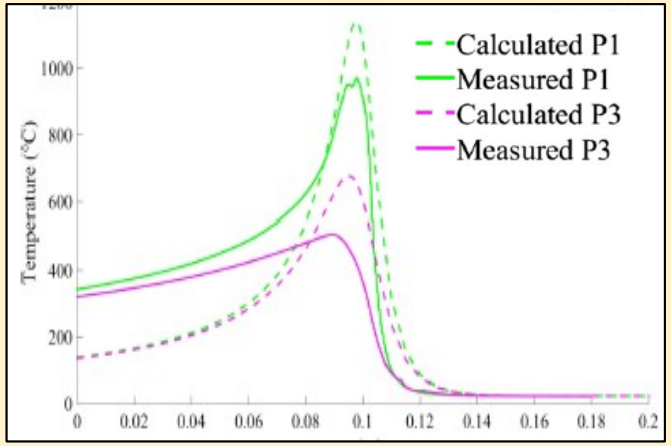
$$T = T_0 + \frac{1}{\sqrt{\left(\frac{x}{a_m}\right)^2 + \left(\frac{\xi}{b_m}\right)^2 + \left(\frac{z}{c_m}\right)^2 + {d_m}^2}} e^{(-B(R+\xi))}$$

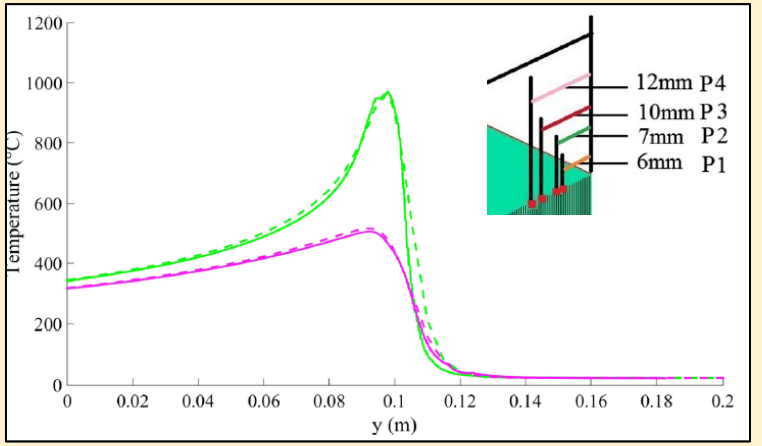
$$a_m = \frac{Q}{2\pi k} \times a; b_m = \frac{Q}{2\pi k} \times b; c_m = \frac{Q}{2\pi k} \times c; d_m$$



Part 02







EX:

Laser Powder Bed Fusion (L PBF)

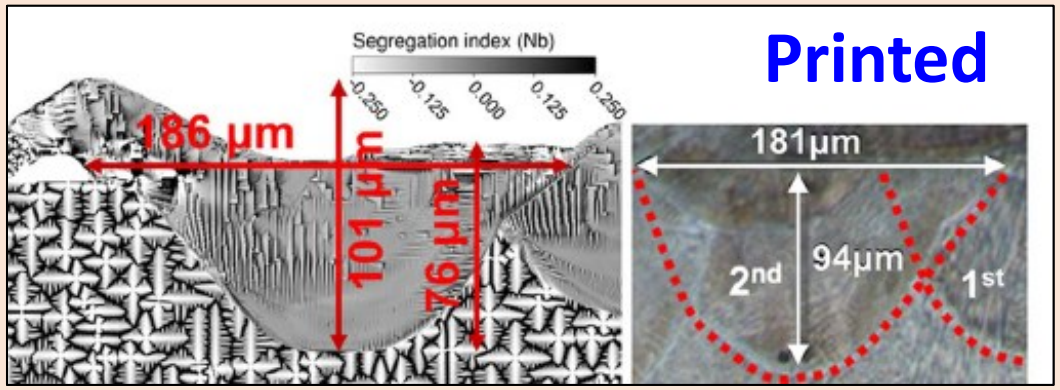
nature communications

Article

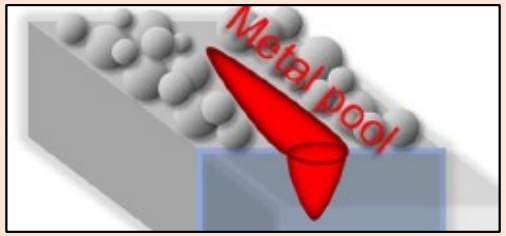
https://doi.org/10.1038/s41467-023-43563-x

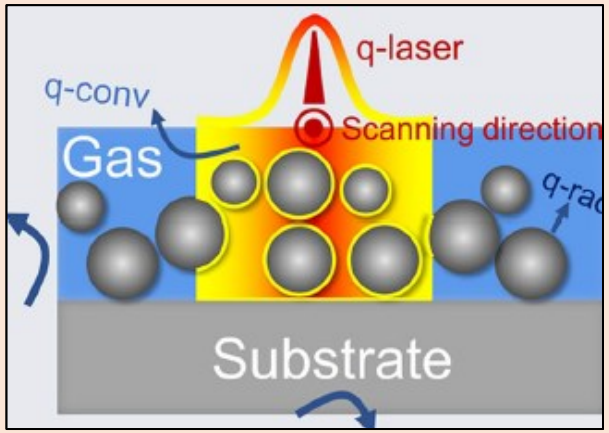
Micro-scale

Solute trapping and non-equilibrium microstructure during rapid solidification of additive manufacturing

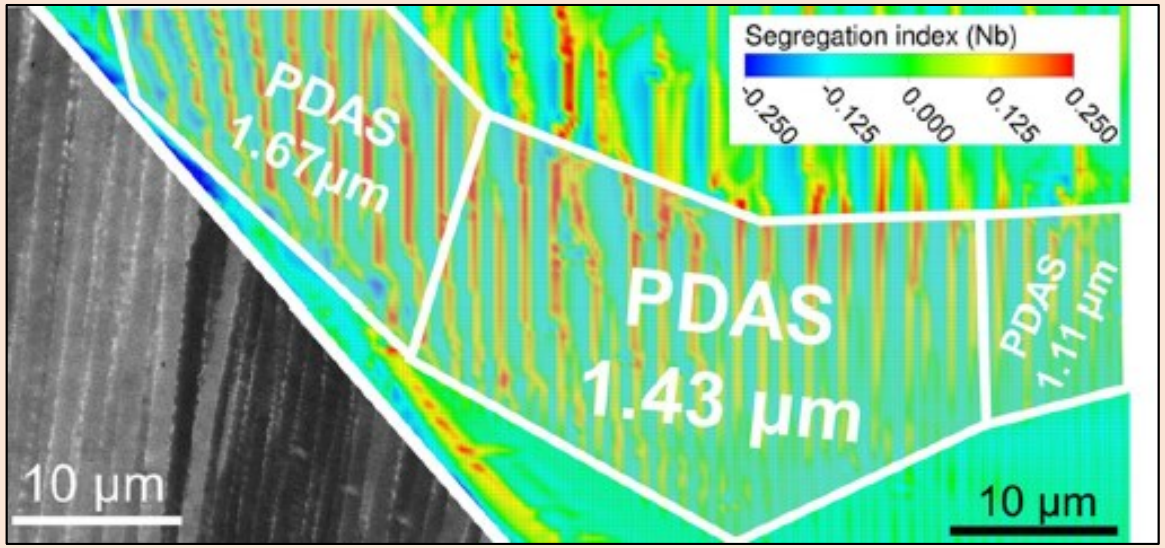


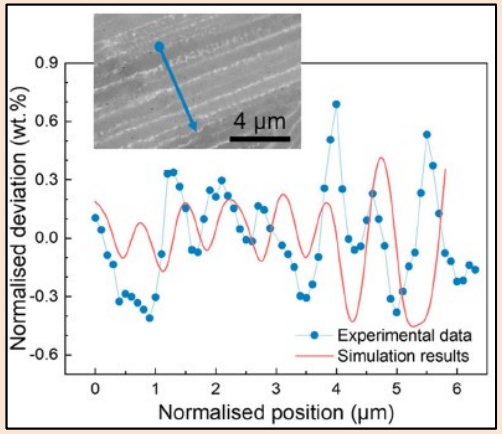
- insight into the solute transport process;
- high-fidelity physics-based numerical modelling;
- effect of melt convection and solute partition.





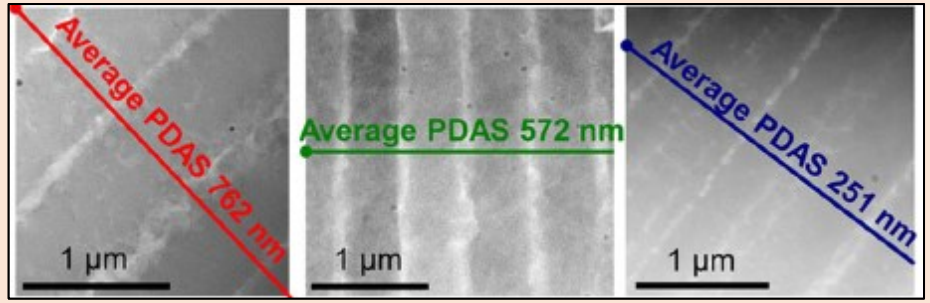
Model





Temperature, flow velocity, solute concentration, and microstructure

... directly coupled



210 × 400 µm computational domain (2D) 1,341,561 grid 0.25 µm

EX:

A Chimera method for thermal part-scale metal Additive Manufacturing simulation

Substepped and advected subdomain methods for part-scale LPBF modeling

Heat Affected Zone (HAZ)

& FEniCS

FEniCS

multi-timestep method

the balance of energy equation

+ phase change

+ growing domain

born-dead element technique

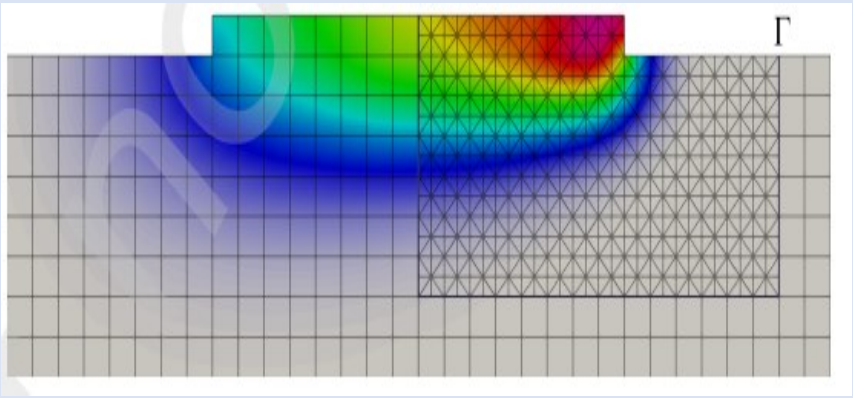
Simulação do processo

Digital twin

moving mesh

moving heat source

Adaptive Mesh Refinement



$$\rho c_p \partial_t T(x,t) - \nabla \cdot \left(k \nabla T(x,t)\right) = r(x,t)$$

$$\rho c_p \left(\partial_t T^m(\xi,t) - \mathbf{v}(t) \cdot \nabla T^m(\xi,t)\right) - \nabla \cdot \left(k \nabla T^m(\xi,t)\right) = r(\xi)$$

ρ, c_p

1 density

2 specific and latent heats

ρ, ξ

1 heat source volumetric

2 New coordinate system

v(t) Add advection term to account for the motion of the mesh.

docker

https://github.com/ordinary-slim/moving_heat_source

https://github.com/ordinary-slim/mhs_fenicsx

$$\nabla . \mathbf{V} = 0$$ Continuidade

$$\rho \frac{\partial \mathbf{V}}{\partial t} = \rho \mathbf{g} - \nabla \mathbf{p} + \mu \nabla^2 \mathbf{V}$$ Qde de movimento

$$\rho C_V \frac{\partial T}{\partial t} = K \nabla^2 T + \Phi$$ Energia

Solving PDEs in Python



Partial Differential Equations

https://github.com/grossiRM/CFD_ON/blob/main/04_OpenFoam/FEniCS/01b__A%20FEniCS%20Tutorial__ii__.pdf